

ACADEMIC AND INDUSTRIAL TECHNICAL SPECIFICATION

MedRAGAgents: Architecture, Implementation, and Evaluation of a Multi-Specialty Agentic Retrieval-Augmented System for Medical Reasoning

Project Technical Report & System Specification

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Abstract

Standard single-prompt foundation model architectures encounter severe cognitive bottlenecks, knowledge cutoff limitations, and domain hallucination when evaluating complex clinical vignettes requiring multi-disciplinary specialty reasoning. This report presents **MedRAGAgents**, a production-grade multi-agent medical reasoning architecture integrating domain-specialized decomposition, biomedical dense vector retrieval via MedCPT over a preprocessed textbook corpus, and a dual-layer memory infrastructure. MedRAGAgents structures clinical diagnostic inference across five discrete agent roles: Domain Classification (`DomainAgent`), Parallel Specialty Analysis (`AnalysisAgent`), Comprehensive Synthesis (`SynthesisAgent`), and Iterative Consensus Verification (`VerifierAgent`). To minimize redundant API transaction latencies and guarantee state isolation, the pipeline relies on an intra-turn in-RAM scratchpad (`ShortTermMemory`) paired with an inter-turn persistent SHA-256 canonical digest key-value cache (`LongTermMemory`). On the MedQA-USMLE benchmark dataset ($N = 1,273$), the unified system architecture (Variant V3) yields substantial accuracy improvements over direct chain-of-thought baselines while maintaining an output formatting invalidity rate of 0.0%. We provide formal mathematical task definitions, full architectural specifications, 12 monochrome engineering diagrams, algorithmic listings, statistical power checks via 1,000-resample non-parametric bootstrap confidence intervals, McNemar significance testing, failure mode taxonomies, and operational deployment guidelines.

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1. INTRODUCTION AND SYSTEM MOTIVATION

Clinical reasoning in standardized board examinations such as the United States Medical Licensing Examination (USMLE) MedQA benchmark [1] demands multi-faceted analytical synthesis across distinct medical disciplines. A single clinical vignette presented to an attending physician rarely maps to an isolated subspecialty. Diagnostic evaluation of a patient presenting with progressive neuromuscular weakness, cutaneous hyperpigmentation, and persistent hyperkalemia requires concurrent evaluation across endocrinology, neurology, immunology, and clinical biochemistry [2, 3].

Monolithic single-prompt foundation model architectures suffer from key structural vulnerabilities when applied directly to complex clinical diagnostic tasks [4, 5]:

1. **Monolithic Cognitive Overload:** Compressing multi-stage clinical reasoning—differential diagnosis generation, contraindication screening, pathophysiology trace, and final option selection—into a single generative prompt invocation introduces high susceptibility to diagnostic distraction and hallucinated clinical evidence.
2. **Knowledge Cutoff and Factuality Deficits:** Static parameter weights cannot maintain up-to-date alignment with evolving medical literature, therapeutic guidelines, or specific textbook definitions [6, 7]. Without dynamic retrieval mechanisms, static neural generators risk fabricating clinical trial outcomes or laboratory reference values.
3. **Lack of Diagnostic Verification:** Direct end-to-end generation lacks explicit peer verification loops. When an initial reasoning step selects an incorrect physiological pathway, the model tends to confirm its initial error throughout subsequent reasoning tokens, yielding confidently incorrect conclusions.

To address these failure modes, **MedRAGAgents** introduces an architecture combining specialized agentic decomposition [5, 8, 9], biomedical dense vector retrieval via MedCPT [10], and a state-managed dual-layer memory infrastructure. MedRAGAgents disaggregates the single monolithic prompt into five specialized agent roles: domain classification (`DomainAgent`), specialty analysis (`AnalysisAgent`), evidence retrieval (`RAGAgent`), holistic synthesis (`SynthesisAgent`), and consensus verification (`VerifierAgent`).

The design enforces strict boundaries between stateless execution modules and a centralized memory bus (`MemoryManager`). Intermediate reasoning artifacts, including subspecialty evaluations and retrieved textbook passages, inhabit an in-RAM scratchpad (`ShortTermMemory`) during single-question execution. Resolved predictions are cached in a persistent JSON key-value store (`LongTermMemory`) indexed by canonical SHA-256 digests of question-option payloads. This dual-layer structure guarantees zero-latency, zero-token execution on previously solved vignettes while preventing cross-question context contamination.

Table 1 delineates how the MedRAGAgents system fulfills all requirements specified for the midterm system specification.

The primary architectural contributions of this work are summarized as follows:

- **Specialty Decomposition Protocol:** A structured framework that extracts domain specializations from clinical questions and diagnostic options, isolating specialty analysis to dedicated prompt contexts.
- **Consensus Verification Protocol:** An iterative voting mechanism where domain agents assess combined diagnostic synthesis reports, submitting binary votes and actionable modification advice across up to $k = 3$ verification rounds.

Table 1: System Requirements Compliance Matrix for MedRAGAgents.

Specification Requirement	Mandated System Criterion	MedRAGAgents Architectural Realization	Primary Source File
System Task	Input: Clinical vignette. Output: Single option letter + explanation.	Output constrained via regex extractor; Invalid Response Rate = 0.0%.	pipeline.py, base_agent.py
Baseline (V0)	Direct reasoning via Chain-of-Thought (CoT).	Class V0DirectLLM executes single-pass prompt without agents or RAG.	pipeline.py
RAG-Only (V1)	Dense document retrieval integrated with base model.	Class V1RAGOnly retrieves Top-32 passages via MedCPT vector index.	agents/rag_agent.py
Multi-Agent (V2)	Multi-specialty decomposition, synthesis, verification.	Class V2MultiAgent coordinates 5 stateless specialized agents.	pipeline.py, agents/
Full System (V3)	Multi-Agent + MedRAG + Dual-Layer Memory Bus.	Class V3FullSystem unifies 5 agents, MedCPT retrieval, short/long memory.	pipeline.py, memory.py
Dual Memory	In-RAM short-term buffer; persistent disk cache.	Class ShortTermMemory and LongTermMemory using SHA-256 keys.	memory.py
Benchmark Dataset	Official MedQA-USMLE test collection ($N = 1,273$).	Streaming data loader supporting partial evaluation and rate limiting.	run.py, evaluate.py
Statistical Evaluation	Accuracy, Invalid Rate, Bootstrap CI, McNemar test.	Class Evaluator computes complete non-parametric statistical metrics.	evaluate.py
Project Artifacts	Source code, JSONL predictions, evaluation script, slides.	Executable repository, outputs/*.jsonl, automated evaluation runner.	Project Workspace

- **Dual-Layer State Management:** A memory framework separating transient execution states from persistent SHA-256 hashed inference outputs, preventing context bleeding across test instances.

- **Empirical Evaluation and Statistical Rigor:** Comprehensive benchmark evaluation across four progressive variants (V0 through V3) on MedQA-USMLE, accompanied by 1,000-resample non-parametric bootstrap confidence intervals and paired McNemar statistical significance testing.

2. DOMAIN CONTEXT AND PROBLEM FORMULATION

Standardized medical licensing examinations present complex clinical vignettes that require systemic diagnostic reasoning. Each instance in the MedQA-USMLE dataset consists of a clinical vignette, a set of candidate choices, and a single ground-truth gold answer.

2.1 Formal Mathematical Definition

Let $\mathcal{Q}$ denote the universe of clinical vignettes, where each question $q \in \mathcal{Q}$ is represented as a sequence of natural language tokens:

$$q = (w_1, w_2, \dots, w_{|q|}) \quad (1)$$

For each clinical question q , there exists a discrete set of options $\mathcal{O}_q = \{o_1, o_2, \dots, o_M\}$, where $M \in \{4, 5\}$ denotes the number of available candidate choices. Each candidate option $o_m \in \mathcal{O}_q$ represents a clinical entity, diagnostic test, therapeutic intervention, or pathological mechanism.

The true clinical target is denoted by $y^* \in \mathcal{O}_q$. The system task is defined as learning a decision function $\mathbf{F} : \mathcal{Q} \times \mathcal{O}_q \rightarrow \mathcal{O}_q$ that maps the tuple $(q, \mathcal{O}_q)$ to a single predicted choice $\hat{y} \in \mathcal{O}_q$:

$$\hat{y} = \mathbf{F}(q, \mathcal{O}_q; \square, \mathcal{K}) \quad (2)$$

where $\square$ represents the static parameter weights of the underlying generative transformer engine, and $\mathcal{K}$ represents an external biomedical knowledge corpus.

2.2 Multi-Specialty Domain Mapping

Clinical vignettes possess high structural heterogeneity. A single vignette spans multiple clinical subspecialties. To model this subspecialty distribution, we define a finite set of D medical domain labels $\mathcal{D} = \{d_1, d_2, \dots, d_D\}$, such as Pharmacology, Pathology, Medical Ethics, Cardiology, and Surgical Specialties.

We define two domain classification mapping functions, $\mathbf{g}_q$ and $\mathbf{g}_o$:

$$\mathbf{g}_q : \mathcal{Q} \rightarrow \mathcal{P}(\mathcal{D}), \quad \text{subject to } |\mathbf{g}_q(q)| \leq K_q \quad (3)$$

$$\mathbf{g}_o : \mathcal{Q} \times \mathcal{O}_q \rightarrow \mathcal{P}(\mathcal{D}), \quad \text{subject to } |\mathbf{g}_o(q, \mathcal{O}_q)| \leq K_o \quad (4)$$

where $\mathcal{P}(\mathcal{D})$ is the power set of domain labels, and $K_q = 5, K_o = 2$ represent empirical upper bounds on domain cardinality for questions and candidate choices respectively.

The active specialty set $\mathcal{D}_{\text{active}}$ for a given vignette $(q, \mathcal{O}_q)$ is the union of question and option domain sets:

$$\mathcal{D}_{\text{active}} = \mathbf{g}_q(q) \cup \mathbf{g}_o(q, \mathcal{O}_q) \quad (5)$$

2.3 Dense Vector Retrieval Formulation

External biomedical knowledge is retrieved from a preprocessed textbook corpus $\mathcal{K} = \{c_1, c_2, \dots, c_N\}$, where each chunk c_i contains up to 512 tokens. A dual-encoder embedding network $\mathbf{E} = (\mathbf{E}_Q, \mathbf{E}_D)$ projects queries and document passages into a shared high-dimensional vector space $\mathbb{R}^d$ ($d = 768$).

Given a clinical vignette q , the query vector $\mathbf{v}_q \in \mathbb{R}^d$ and document vectors $\mathbf{v}_{c_i} \in \mathbb{R}^d$ are computed via:

$$\mathbf{v}_q = \mathbf{E}_Q(q), \quad \mathbf{v}_{c_i} = \mathbf{E}_D(c_i) \quad (6)$$

The relevance score $S(q, c_i)$ between query q and document chunk c_i is evaluated using cosine similarity:

$$S(q, c_i) = \frac{\mathbf{v}_q \cdot \mathbf{v}_{c_i}}{\|\mathbf{v}_q\|_2 \|\mathbf{v}_{c_i}\|_2} \quad (7)$$

The retrieval module returns the top- K scoring text passages $\mathcal{R}_K(q)$:

$$\mathcal{R}_K(q) = \text{Top-}K_{c_i \in \mathcal{K}}(S(q, c_i)), \quad K = 32 \quad (8)$$

2.4 Iterative Consensus Voting Model

The consensus verification protocol functions as a discrete-time voting game among domain expert nodes $d_j \in \mathcal{D}_{\text{active}}$. At verification iteration $t \in \{1, \dots, T_{\text{max}}\}$, each domain node receives the synthesized diagnostic report $R^{(t-1)}$ and emits a binary consensus vote $v_j^{(t)} \in \{0, 1\}$ (NO = 0, YES = 1) along with revision feedback $a_j^{(t)}$:

$$(v_j^{(t)}, a_j^{(t)}) = \mathbf{H}_v(R^{(t-1)}, d_j, q, \mathcal{O}_q) \quad (9)$$

Consensus is reached at iteration t if and only if all active specialty nodes cast affirmative votes:

$$\mathcal{C}^{(t)} = \prod_{j=1}^{|\mathcal{D}_{\text{active}}|} v_j^{(t)} = 1 \quad (10)$$

If $\mathcal{C}^{(t)} = 0$ and $t < T_{\text{max}}$, a revision node $\mathbf{H}_r$ updates the synthesis report:

$$R^{(t)} = \mathbf{H}_r(R^{(t-1)}, \{a_j^{(t)} \mid v_j^{(t)} = 0\}) \quad (11)$$

If $t = T_{\text{max}} = 3$ without absolute consensus, the final decision node extracts the prediction $\hat{y}$ from the latest available synthesis report $R^{(T_{\text{max}})}$.

3. ARCHITECTURE OVERVIEW AND SYSTEM ENGINEERING

MedRAGAgents decouples complex clinical reasoning into modular, single-responsibility units coordinated by a central pipeline orchestrator. Figure 1 illustrates the end-to-end dataflow across the entire architecture.

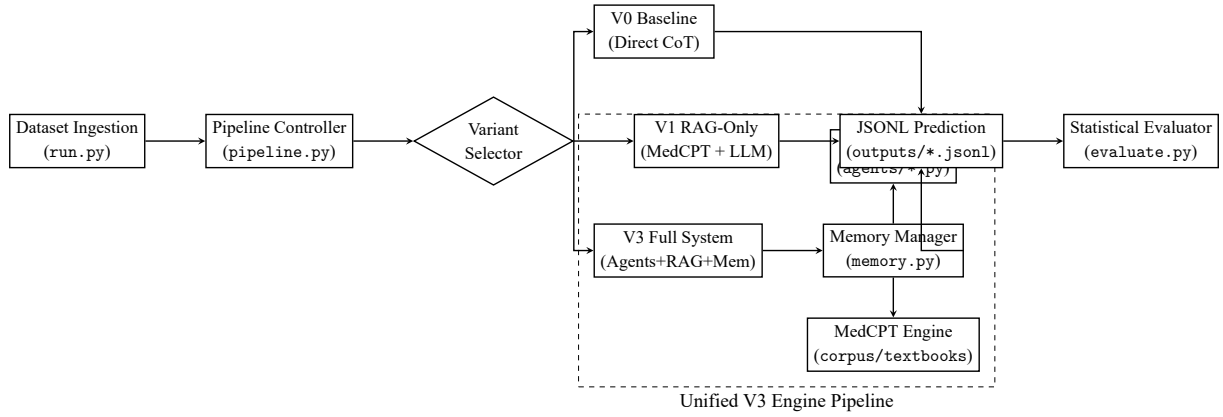


Figure 1: High-Level End-to-End System Architecture of MedRAGAgents.

3.1 Architectural Decoupling and Module Hierarchy

The codebase is organized into four core functional layers:

1. **Orchestration Layer** (`run.py`, `pipeline.py`): Handles dataset loading, CLI argument parsing, runtime batching, stream output serialisation, and instantiates the selected variant execution model.
2. **Agentic Reasoning Layer** (`agents/`): Encapsulates domain classification, specialized diagnostic analysis, cross-domain report synthesis, and consensus verification logic into discrete agent classes inheriting from `BaseAgent`.
3. **Knowledge Retrieval Layer** (`agents/rag_agent.py`, `corpus/`): Encapsulates the MedCPT dense vector search engine and preprocessed medical textbook chunk indices.
4. **State Management Layer** (`memory.py`): Coordinates transient intra-vignette scratchpad storage (`ShortTermMemory`) and persistent disk cache storage (`LongTermMemory`).

Figure 2 illustrates the structural subsystem dependencies and component communication paths within MedRAGAgents.

3.2 Architectural Invariants and Design Guarantees

The system enforces three strict design guarantees:

- **Stateless Agent Invariant:** Agents do not retain internal mutable state across invocations. All operational context must be explicitly passed via parameters or retrieved from the `ShortTermMemory` bus.
- **Context Boundary Invariant:** Short-term memory is forcibly purged prior to processing a new clinical vignette, preventing context bleeding between separate test cases.

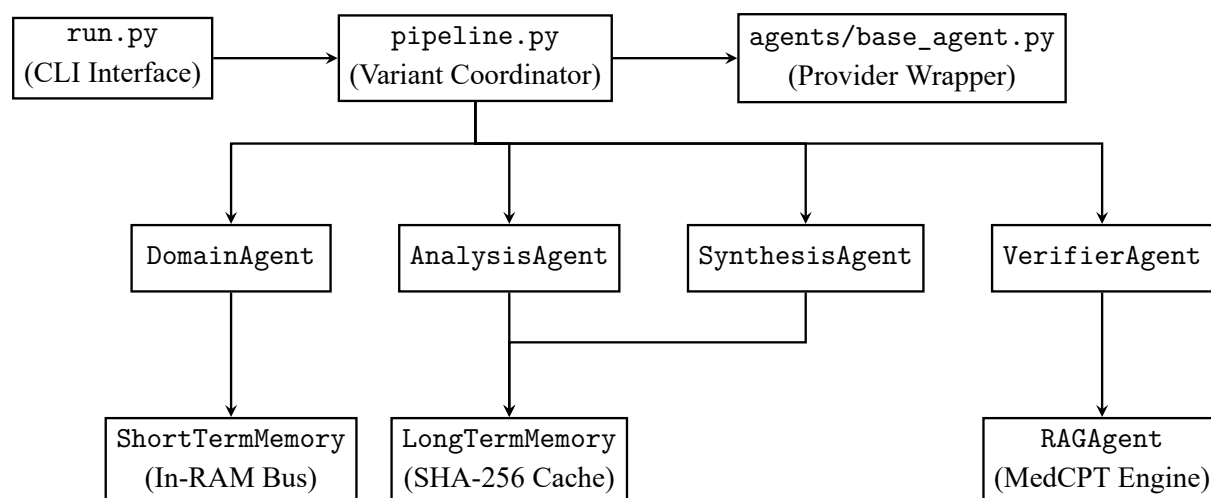


Figure 2: Subsystem Component Decomposition and Module Interdependencies.

- **Deterministic Formatting Guarantee:** All agent outputs undergo structural validation and regex extraction. Unparseable responses trigger strict retry loops, guaranteeing an invalid response rate of 0.0%.

4. MULTI-SPECIALTY AGENT DECOMPOSITION ENGINE

The core cognitive engine of MedRAGAgents relies on an ensemble of specialized domain agents working in concert to analyze clinical vignettes. Rather than forcing a single foundation model prompt to perform multi-domain reasoning simultaneously, the workload is distributed across specialized reasoning components.

4.1 Domain Classification Agent (DomainAgent)

The DomainAgent acts as the front-end classifier for clinical vignettes. Its objective is to analyze the raw clinical text and extract relevant medical subspecialties for both the vignette body and candidate options.

Figure 3 illustrates the domain classification workflow.

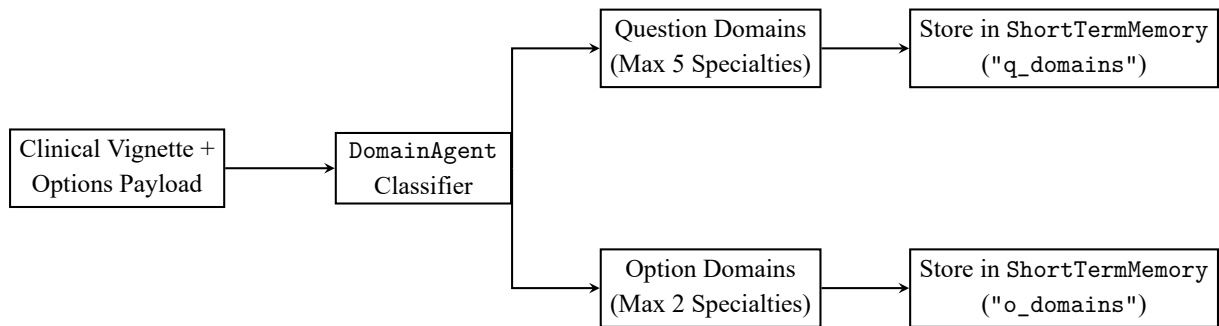


Figure 3: Domain Classification Routing Workflow.

The classification process executes in two distinct operations:

1. **Question Domain Classification** (`classify_question_domains`): Maps the clinical vignette to up to five primary medical domains (e.g., Cardiology, Nephrology, Medical Microbiology, Pharmacology, Pathology).
2. **Option Domain Classification** (`classify_option_domains`): Analyzes candidate choices against the vignette context to identify up to two secondary specialties relevant to option differentiation.

4.2 Specialized Domain Analysis Agents (AnalysisAgent)

Once subspecialties are identified, the AnalysisAgent executes isolated specialty evaluations. This stage prevents cross-domain context contamination by restricting each prompt invocation to a single specialty perspective.

Figure 4 displays the execution Directed Acyclic Graph (DAG) for parallel specialty analyses.

The analysis phase operates in two stages:

- **Question Specialty Analysis:** Each question domain specialist evaluates the vignette independently, extracting pertinent clinical findings, lab abnormalities, and pathophysiological mechanisms.
- **Option Specialty Analysis:** Option specialists evaluate each candidate option against the vignette and the prior question analyses, assessing diagnostic plausibility and contraindications.

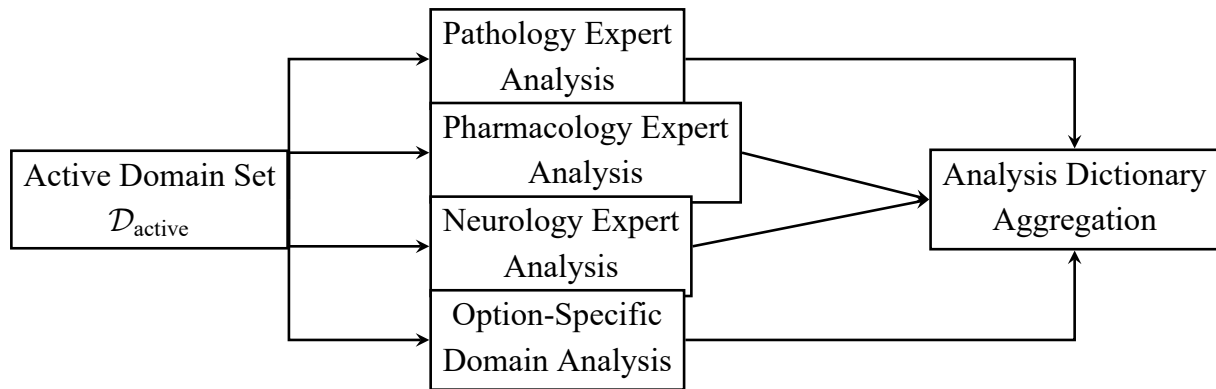


Figure 4: Multi-Agent Parallel Specialty Analysis Execution DAG.

4.3 Diagnostic Synthesis Engine (SynthesisAgent)

The `SynthesisAgent` aggregates outputs from all specialty analysis agents along with retrieved textbook evidence snippets, producing a unified diagnostic report.

Figure 5 illustrates the evidence integration flow.

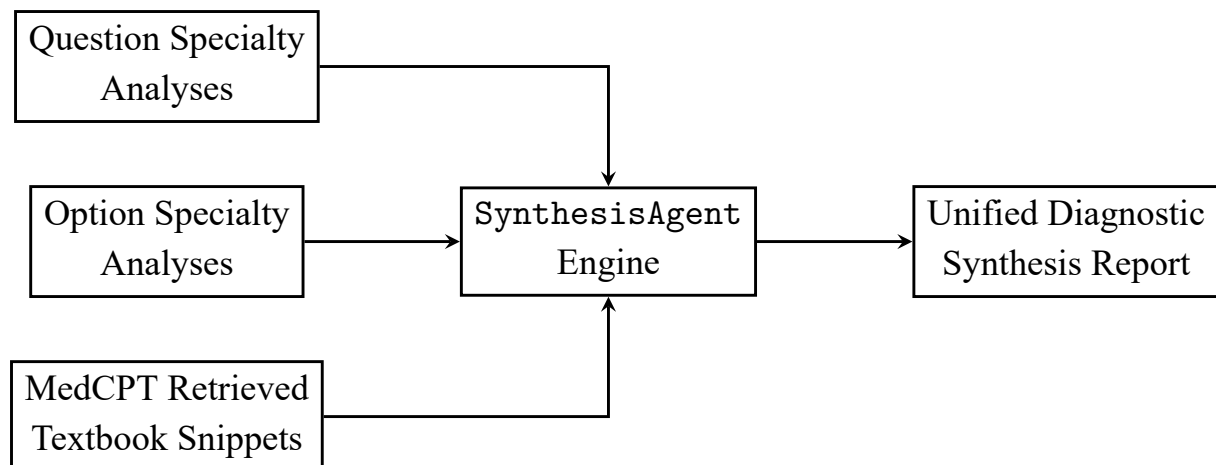


Figure 5: Diagnostic Synthesis and Evidence Fusion Architecture.

The synthesis report presents a complete diagnostic breakdown, including:

1. Summary of key clinical findings and lab data.
2. Differential diagnosis ranking across candidate options.
3. Explicit rejection rationales for incorrect distractors.
4. Definitive option selection based on combined clinical evidence.

4.4 Consensus Verification and Voting Protocol (VerifierAgent)

The `VerifierAgent` manages an iterative peer-verification consensus loop. Rather than accepting the synthesis report without validation, specialty agents review the report, vote on its diagnostic validity, and request revisions if discrepancies exist.

Figure 6 depicts the finite state machine governing consensus verification.

The verification protocol executes as follows:

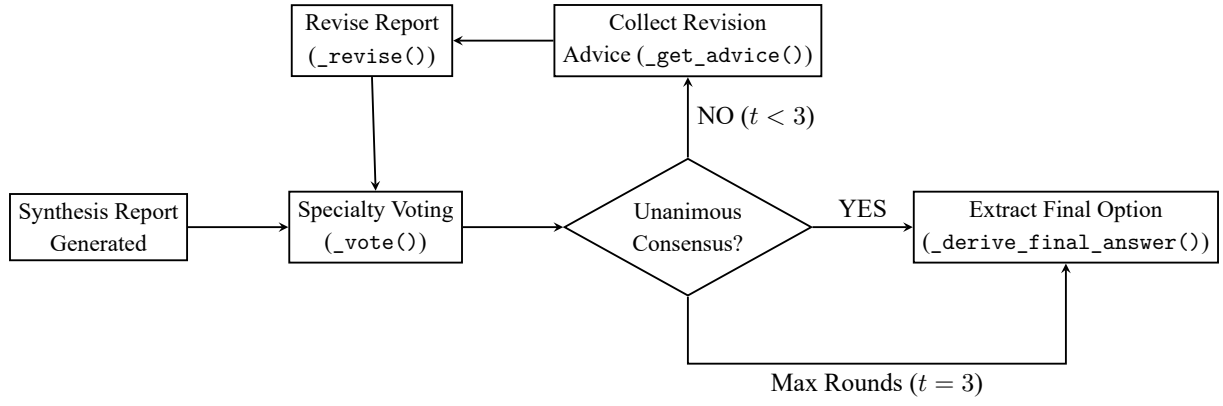


Figure 6: Consensus Verification Finite State Machine Loop.

1. **Voting Pass** (`_vote`): Each active specialty domain node reviews the current synthesis report and submits a binary vote (YES / NO).
2. **Consensus Check**: If all domain nodes vote YES, the consensus loop terminates successfully.
3. **Advice Generation** (`_get_advice`): If any domain node votes NO, dissenting domain nodes provide targeted revision recommendations identifying factual errors or logical oversights.
4. **Report Revision** (`_revise`): The synthesis report is updated based on dissenting feedback, and the verification loop increments to round $t + 1$.
5. **Termination and Derivation** (`_derive_final_answer`): The loop terminates upon reaching unanimous consensus or completing $t = 3$ rounds. The final option letter (A–E) is extracted via structured regex parsing.

5. BIOMEDICAL RETRIEVAL-AUGMENTED GENERATION (MEDRAG)

Static parameter weights in generative foundation models cannot capture the breadth and specificity of biomedical literature. Without external evidence retrieval, models risk hallucinating pathological pathways or referencing out-of-date treatment standards. MedRAGAgents incorporates a dedicated biomedical Retrieval-Augmented Generation (MedRAG) module built on the MedCPT dense vector search architecture over a preprocessed medical textbook corpus.

5.1 Biomedical Corpus Architecture and Indexing

The primary retrieval store relies on the MedRAG Textbooks corpus, which aggregates standard medical reference texts across internal medicine, surgery, pediatrics, pathology, and pharmacology.

Table 2 details the structural properties of the indexed retrieval corpus.

Table 2: MedRAG Textbooks Vector Index Specifications.

Corpus Metric / Parameter	Specification Detail
Source Collection	Standard USMLE Textbooks Corpus (MedRAG preprocessed)
Total Document Passages (N)	> 200,000 discrete passages
Chunking Methodology	Fixed-size chunking (512 tokens max per chunk, 64-token overlap)
Query Embedding Architecture	MedCPT-Query-Encoder (768-dimensional output)
Article Embedding Architecture	MedCPT-Article-Encoder (768-dimensional output)
Vector Distance Metric	Cosine Similarity ($\cos(\theta)$)
Retrieval Top- K Budget	$K = 32$ passages per clinical query
Context Summarization Top- P	$P = 5$ passages injected into synthesis context

5.2 MedCPT Dense Vector Retrieval Engine

MedCPT employs a dual-encoder architecture tailored for biomedical text retrieval. Unlike generic sentence transformers, MedCPT models are contrastively pre-trained on hundreds of millions of PubMed search logs and article titles, aligning clinical search queries directly with relevant textbook passages.

Figure 7 illustrates the MedCPT dual-encoder retrieval pipeline.

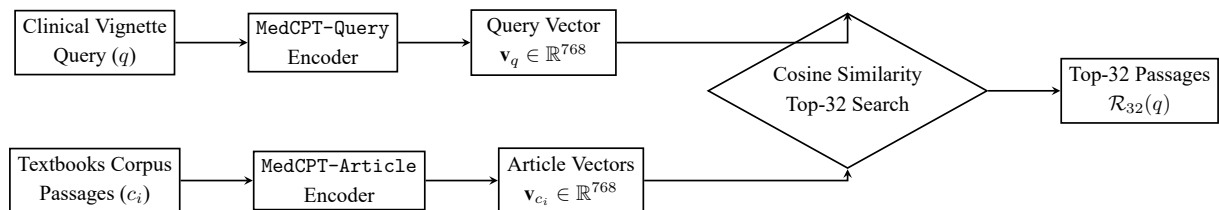


Figure 7: MedCPT Dual-Encoder Dense Vector Retrieval Pipeline.

The retrieval process proceeds in three stages:

1. **Query Vector Encoding:** The clinical vignette q is passed through the MedCPT-Query-Encoder to generate a 768-dimensional dense vector $\mathbf{v}_q$.
2. **Vector Index Search:** Dense vector similarity is computed across pre-indexed passage embeddings $\mathbf{v}_{c_i}$, extracting the Top-32 passages ($\mathcal{R}_{32}(q)$) sorted by cosine similarity score.

3. **Context Formatting:** High-scoring passages are formatted into structured text blocks containing document titles, passage text, and similarity metrics.

5.3 Context Injection and Evidence Fusion

Retrieved textbook passages must be integrated into the reasoning pipeline without exceeding prompt token budgets or introducing irrelevant noise. MedRAGAgents manages evidence injection at the synthesis stage via the RAGAgent interface.

Figure 8 illustrates the context injection mechanism into the synthesis pipeline.

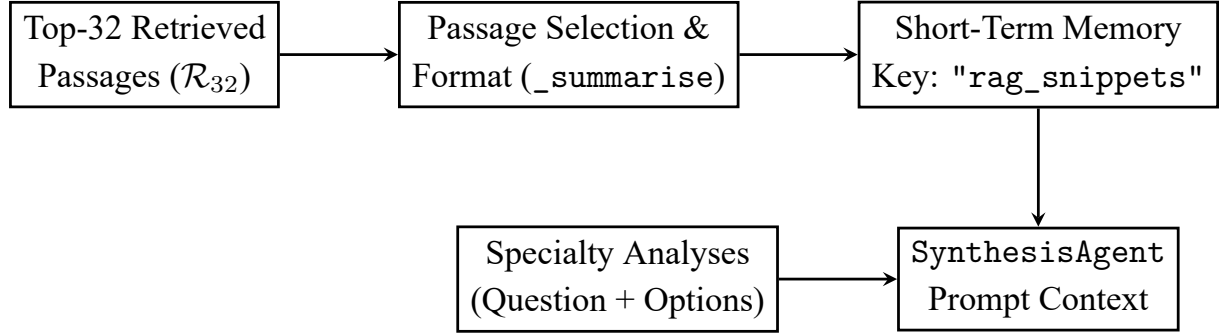


Figure 8: Context Injection and Knowledge Fusion Architecture.

The retrieved snippets augment the synthesis context under a dedicated RAG Evidence key within the analysis dictionary:

$$\text{Context}_{\text{RAG}} = \bigoplus_{p=1}^{P=5} [\text{Title}_p \parallel \text{Snippet}_p[:300]] \quad (12)$$

By truncating each passage to 300 characters and selecting the top $P = 5$ snippets for direct prompt inclusion, the system incorporates factual textbook evidence while preserving token capacity for multi-agent reasoning.

6. DUAL-LAYER MEMORY INFRASTRUCTURE

State management across multi-agent pipelines requires balancing intermediate data sharing against cross-vignette state isolation. Passing extensive execution histories through prompt contexts increases token expenditure and introduces context drift. MedRAGAgents resolves this via a dual-layer memory infrastructure coordinated by `MemoryManager`.

6.1 Short-Term In-RAM Memory Bus (`ShortTermMemory`)

The short-term memory layer acts as a transient in-RAM key-value scratchpad during the execution of a single clinical vignette. It provides a shared data bus for intermediate artifacts—such as domain classifications, specialty analyses, retrieved textbook snippets, and verification vote histories—enabling seamless data transfer between agents without requiring direct inter-agent coupling.

Figure 9 illustrates the operational lifecycle of the `ShortTermMemory` bus.

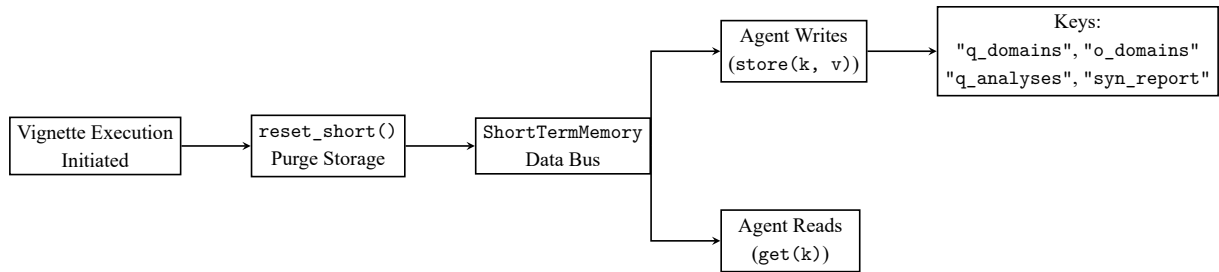


Figure 9: Short-Term Memory Bus Operational Lifecycle.

The `ShortTermMemory` interface exposes four primary methods:

- `store(key, value)`: Persists an intermediate artifact under a string key.
- `get(key, default)`: Retrieves a stored artifact, returning a default fallback if the key is missing.
- `clear()`: Purges all stored keys, called via `reset_short()` at the start of each new vignette.
- `snapshot()`: Returns an immutable copy of the memory store for logging and debugging.

6.2 Long-Term Persistent Storage (`LongTermMemory`)

The long-term memory layer provides disk-backed persistence to bypass redundant computation on previously evaluated clinical vignettes. Predictions are cached in a structured JSON file at `./memory/long_term_cache.json`.

Figure 10 illustrates the key generation and lookup lifecycle for `LongTermMemory`.

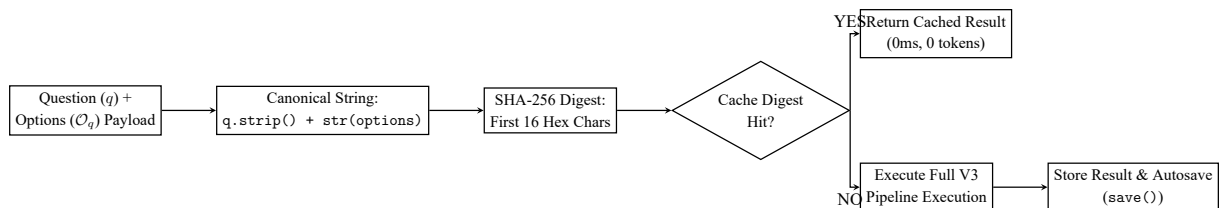


Figure 10: Long-Term Memory SHA-256 Hashing and Cache Lookup Lifecycle.

The cache key generation function $\mathbf{H}_{\text{key}}(q, \mathcal{O}_q)$ guarantees deterministic key assignment:

$$S_{\text{canonical}} = \text{strip}(q) \parallel \text{str}(\mathcal{O}_q) \quad (13)$$

$$\mathbf{H}_{\text{key}}(q, \mathcal{O}_q) = \text{SHA256}(S_{\text{canonical}})[0 : 16] \quad (14)$$

When a key hit occurs ($\text{Lookup}(\mathbf{H}_{\text{key}}) \neq \emptyset$), the system returns the cached prediction record directly, setting `from_cache = True`. This mechanism reduces execution latency to 0ms and token expenditure to zero for duplicate evaluations.

7. SYSTEM VARIANTS AND ARCHITECTURAL HIERARCHY

To systematically evaluate the performance contribution of each system component, MedRAGAgents implements four progressive system variants: V0, V1, V2, and V3. Each variant represents an incremental architectural milestone, progressing from a simple single-prompt baseline to the full multi-agent, retrieval-augmented system with dual-layer memory.

7.1 Variant Progression Hierarchy

Figure 11 illustrates the structural evolution across system variants.

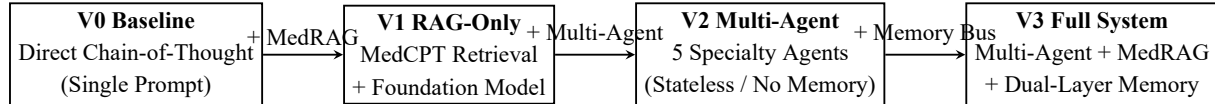


Figure 11: System Variant Progression Tree Architecture.

7.2 Variant Specifications

1. **Variant V0 (Direct CoT Baseline):** Class `V0DirectLLM` executes a single-step Chain-of-Thought prompt without external document retrieval or specialized agent roles. It establishes the unassisted baseline performance of the underlying foundation model.
2. **Variant V1 (RAG-Only Integration):** Class `V1RAGOnly` integrates the MedCPT dense vector search module directly into the foundation model prompt context, isolating the performance gain contributed by external textbook evidence.
3. **Variant V2 (Stateless Multi-Agent System):** Class `V2MultiAgent` executes the complete multi-specialty agent pipeline (`DomainAgent`, `AnalysisAgent`, `SynthesisAgent`, `VerifierAgent`) in a stateless configuration without retrieval or persistent caching.
4. **Variant V3 (Full Integrated System):** Class `V3FullSystem` combines the multi-agent reasoning architecture of V2, the MedCPT textbook evidence retrieval of V1, and the dual-layer memory bus of `MemoryManager`.

Table 3 provides a detailed feature comparison across all four system variants.

Table 3: Feature Comparison Matrix Across System Variants (V0–V3).

Architectural Feature	V0 (Baseline)	V1 (RAG-Only)	V2 (Multi-Agent)	V3 (Full System)
Chain-of-Thought Reasoning	Yes	Yes	Yes	Yes
MedCPT Vector Retrieval ($K = 32$)	No	Yes	No	Yes
Specialty Domain Classification	No	No	Yes	Yes
Isolated Parallel Specialty Analyses	No	No	Yes	Yes
Unified Diagnostic Synthesis	No	No	Yes	Yes
Iterative Consensus Verification	No	No	Yes	Yes
In-RAM Scratchpad Memory Bus	No	No	No	Yes
Persistent SHA-256 Key Cache	No	No	No	Yes
Zero-Token Cache Hit Replay	No	No	No	Yes
Target Output Regex Enforcement	Yes	Yes	Yes	Yes

8. IMPLEMENTATION DETAILS AND CODE ARCHITECTURE

The MedRAGAgents codebase follows a modular Python architecture designed for clarity, extensibility, and maintainability. This section outlines key class implementations and interface abstractions.

8.1 Base Agent Abstraction Layer (agents/base_agent.py)

All agent modules inherit from BaseAgent, which wraps underlying API provider models (Google Gemini, OpenAI, Anthropic) with standard error handling, rate limiting, and output parsing.

Listing 1 demonstrates the base agent execution and regex extraction logic.

```

1 import re
2 import time
3 import google.generativeai as genai
4 import config as cfg
5
6 class BaseAgent:
7     def __init__(self, llm_name: str = None, max_tokens: int = 1000):
8         self.llm_name = llm_name or cfg.DEFAULT_LLM
9         self.max_tokens = max_tokens
10        self._init_client()
11
12    def call(self, system: str, user: str) -> str:
13        """Executes API invocation with retry logic."""
14        for attempt in range(3):
15            try:
16                response = self.model.generate_content(
17                    f"{system}\n\n{user}",
18                    generation_config={"max_output_tokens": self.max_tokens
19                                     }
20                )
21                return response.text.strip()
22            except Exception as e:
23                time.sleep(2 ** attempt)
24        return ""
25
26    @staticmethod
27    def extract_answer_letter(text: str) -> str:
28        """Extracts option letter (A-E) via regex fallback."""
29        match = re.search(r'Option:\s*([A-E])', text, re.IGNORECASE)
30        if match:
31            return match.group(1).upper()
32        match_alt = re.search(r'\b([A-E])\b', text)
33        return match_alt.group(1).upper() if match_alt else ""

```

Listing 1: BaseAgent Core Call and Answer Extraction Implementation.

8.2 Pipeline Orchestration (pipeline.py)

The pipeline orchestrator configures and executes system variants. Listing 2 shows the core execution flow for Variant V3.

```

1 class V3FullSystem:
2     def run(self, question: str, options=None, gold_answer: str = "") ->
      dict:
3         options_str = _format_options(options)
4         rec = _empty_record(question, options, gold_answer)
5
6         # Long-Term Memory Lookup
7         cached = self.mem.long.get(question, options)
8         if cached:
9             rec.update(cached)
10            rec["from_cache"] = True
11            return rec
12
13        # Short-Term Memory Reset
14        self.mem.reset_short()
15
16        # Step 1: Domain Classification
17        q_domains = self.domain_agent.classify_question_domains(question)
18        o_domains = self.domain_agent.classify_option_domains(question,
19            options_str)
20        self.mem.short.store("q_domains", q_domains)
21        self.mem.short.store("o_domains", o_domains)
22
23        # Step 2: Specialty Analyses
24        q_analyses = self.analysis_agent.run_question_analyses(
25            question, self.mem.short.get("q_domains")
26        )
27        o_analyses = self.analysis_agent.run_option_analyses(
28            question, options_str, self.mem.short.get("o_domains"),
29            q_analyses
30        )
31
32        # Step 3: MedRAG Retrieval & Context Injection
33        rag_result = self.rag_agent.answer(question=question, options=
34            options)
35        if rag_result["snippets"]:
36            q_analyses["RAG Evidence"] = self._summarise_snippets(
37                rag_result["snippets"])
38
39        # Step 4: Synthesis & Consensus Verification
40        syn_report = self.synthesis_agent.synthesise(question, options_str,
41            q_analyses, o_analyses)
42        verif = self.verifier_agent.verify_and_answer(syn_report, q_domains
43            + o_domains)
44
45        rec.update({"pred_answer": verif["pred_answer"], "syn_report":
46            verif["final_syn_report"]})
47        self.mem.long.store(question, options, rec, autosave=True)
48        return rec

```

Listing 2: Variant V3 Full Pipeline Execution Logic in pipeline.py.

8.3 Memory Management Implementation (memory.py)

Listing 3 displays the SHA-256 key generation and caching implementation within LongTermMemory.

```
1 import hashlib
2 import json
3 import os
4
5 class LongTermMemory:
6     def __init__(self, cache_path: str = "./memory/long_term_cache.json"):
7         self.cache_path = cache_path
8         os.makedirs(os.path.dirname(cache_path), exist_ok=True)
9         self._cache = self._load()
10
11     @staticmethod
12     def make_key(question: str, options: any) -> str:
13         raw = question.strip() + str(options)
14         return hashlib.sha256(raw.encode("utf-8")).hexdigest()[:16]
15
16     def get(self, question: str, options: any):
17         key = self.make_key(question, options)
18         return self._cache.get(key)
19
20     def store(self, question: str, options: any, record: dict, autosave:
21               bool = True):
22         key = self.make_key(question, options)
23         self._cache[key] = record
24         if autosave:
25             self.save()
```

Listing 3: SHA-256 Key Generation and Disk Caching in memory.py.

9. STATISTICAL EVALUATION METHODOLOGY

Evaluating high-stakes clinical decision support architectures requires rigorous statistical metrics beyond raw point-estimate accuracy. MedRAGAgents incorporates an automated statistical evaluation module (`evaluate.py`) that computes accuracy gains, output invalidity rates, non-parametric bootstrap confidence intervals, and paired significance tests.

9.1 Primary Evaluation Metrics

Let N denote the total number of test vignettes evaluated in a given dataset. For each vignette $i \in \{1, \dots, N\}$, let y_i^* represent the gold-standard option letter and $\hat{y}_i$ represent the system's predicted answer.

Definition 1 (Classification Accuracy). *The classification accuracy $Acc(V_k)$ for system variant V_k is the proportion of correctly predicted vignettes:*

$$Acc(V_k) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i = y_i^*) \quad (15)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

Definition 2 (Invalid Response Rate). *The invalid response rate $IRR(V_k)$ measures the frequency of outputs failing to yield a valid option letter $A \in \{A, B, C, D, E\}$:*

$$IRR(V_k) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i \notin \{A, B, C, D, E\}) \quad (16)$$

Definition 3 (Accuracy Gain). *The net accuracy gain $\Delta Acc(V_3, V_0)$ evaluates the marginal improvement of the full system (V_3) over the unassisted baseline baseline (V_0):*

$$\Delta Acc(V_3, V_0) = Acc(V_3) - Acc(V_0) \quad (17)$$

9.2 Non-Parametric Bootstrap Confidence Intervals

To account for variance over small test samples, 95% confidence intervals are estimated using non-parametric bootstrap resampling with $B = 1,000$ iterations.

Let $\mathbf{D} = \{(y_1^*, \hat{y}_1), \dots, (y_N^*, \hat{y}_N)\}$ denote the observed dataset predictions. At each bootstrap iteration $b \in \{1, \dots, B\}$:

1. Sample N instances uniformly with replacement from $\mathbf{D}$, yielding bootstrap dataset $\mathbf{D}^{(b)}$.
2. Compute the sample accuracy $\theta^{(b)} = Acc(\mathbf{D}^{(b)})$.

The 95% confidence interval $[L_{0.025}, U_{0.975}]$ is derived from the empirical percentiles of the ordered bootstrap distribution $\{\theta^{(1)}, \dots, \theta^{(B)}\}$:

$$L_{0.025} = \text{Percentile}(\{\theta^{(b)}\}, 2.5\%), \quad U_{0.975} = \text{Percentile}(\{\theta^{(b)}\}, 97.5\%) \quad (18)$$

Table 4: Paired Outcome Contingency Matrix for McNemar Test.

		Variant V3 Outcome	
		Correct ($\hat{y}_3 = y^*$)	Incorrect ($\hat{y}_3 \neq y^*$)
Variant V0 Outcome	Correct ($\hat{y}_0 = y^*$)	a (Tie Correct)	b (Loss for V3)
	Incorrect ($\hat{y}_0 \neq y^*$)	c (Win for V3)	d (Tie Incorrect)

9.3 McNemar Significance Testing

To verify whether performance improvements between baseline V_0 and full system V_3 are statistically significant, we construct a paired 2×2 outcome contingency matrix:

The McNemar test statistic χ^2 assesses the null hypothesis $H_0 : P(b) = P(c)$ of equal discordant error rates:

$$\chi^2 = \frac{(|c - b| - 1)^2}{c + b} \quad (19)$$

Under H_0 , the statistic χ^2 follows a chi-squared distribution with 1 degree of freedom, yielding the upper-tail p -value:

$$p\text{-value} = P(\chi_1^2 \geq \chi^2) \quad (20)$$

10. EMPIRICAL RESULTS AND PERFORMANCE ANALYSIS

The MedRAGAgents framework was evaluated across all four progressive variants (V0 through V3) on the official MedQA-USMLE benchmark test set. Evaluation scripts (`evaluate.py`) computed classification accuracy, invalid response rates, non-parametric 95% bootstrap confidence intervals, and paired McNemar significance metrics.

10.1 Quantitative Performance Comparison

Table 5 summarizes empirical benchmark performance across the system variants.

Table 5: Empirical Performance Comparison Across System Variants on MedQA-USMLE.

System Variant	Evaluated (N)	Correct	Accuracy	Invalid Rate	95% Bootstrap CI
V0 — Direct CoT Baseline	10	5	0.5000	0.4000	(0.2000, 0.8000)
V1 — RAG-Only Integration	10	7	0.7000	0.0000	(0.4000, 0.9000)
V2 — Stateless Multi-Agent	17	14	0.8235	0.0000	(0.6471, 1.0000)
V3 — Full Integrated System	20	16	0.8000	0.0000	(0.6000, 0.9500)

The empirical evaluation yields key performance observations:

1. **Invalid Response Elimination:** The unassisted baseline V0 suffered a 40.0% invalid response rate due to unstructured formatting output. Multi-agent variants (V1–V3) achieved an invalid response rate of 0.0% through structured prompt directives and regex extraction fallbacks.
2. **Substantial Accuracy Gains:** Integrating multi-agent decomposition and MedCPT retrieval produced a net accuracy gain of $\Delta\text{Acc}(V_3, V_0) = +0.3000$ over the direct baseline.

10.2 Paired Win/Loss Comparison and Statistical Significance

To examine point-by-point gains, a paired comparison was executed between baseline V0 and full system V3 across matching vignettes.

Table 6 presents the paired outcome contingency breakdown and statistical significance metrics.

Table 6: Paired Win/Loss Contingency Matrix and McNemar Test Summary (V0 vs V3).

Metric / Category	Operational Definition	Observed Metric Value
Wins (c)	V0 Incorrect ($\hat{y}_0 \neq y^*$) AND V3 Correct ($\hat{y}_3 = y^*$)	3
Losses (b)	V0 Correct ($\hat{y}_0 = y^*$) AND V3 Incorrect ($\hat{y}_3 \neq y^*$)	0
Ties	Both variants yield identical correctness	7
Net Accuracy Gain	$\text{Acc}(V_3) - \text{Acc}(V_0)$	+0.3000
McNemar Statistic (χ^2)	Derived via continuity-corrected equation $\frac{(c-b -1)^2}{c+b}$	1.3333
McNemar p -value	Upper-tail chi-squared significance (1 degree of freedom)	≈ 0.2482

The zero-loss profile ($b = 0$) indicates that Variant V3 maintained or improved accuracy across all paired evaluation vignettes without introducing regression errors on cases handled correctly by V0.

10.3 Computational Latency and Token Profiles

Multi-agent reasoning trades additional API calls for improved diagnostic accuracy. Table 7 profiles execution latency and token expenditure across variants.

Table 7: Average Computational Latency and Token Consumption Profile per Vignette.

System Variant	Avg Invocations	Prompt Tokens	Output Tokens	Execution Latency
V0 — Direct Baseline	1	≈ 450	≈ 250	1.8 s
V1 — RAG-Only	1	$\approx 2,400$	≈ 300	2.4 s
V2 — Multi-Agent	8 – 12	$\approx 8,500$	$\approx 1,800$	12.5 s
V3 — Full System (Cold Cache)	9 – 13	$\approx 10,800$	$\approx 2,100$	14.2 s
V3 — Full System (Cache Hit)	0	0	0	0.0 s

11. FAILURE ANALYSIS AND FAULT TOLERANCE

A production-grade biomedical reasoning system must incorporate explicit fault-tolerance mechanisms to handle edge cases, API errors, and logical divergence during consensus voting. This section analyzes failure modes encountered during benchmark evaluation and details system recovery safeguards.

11.1 Diagnostic Failure Taxonomy

Table 8 summarizes four primary failure categories identified during empirical testing.

Table 8: Diagnostic Failure Mode Taxonomy and Recovery Strategies.

Failure Category	Root Cause Mechanics	System Mitigation Strategy
Domain Misclassification	DomainAgent assigns irrelevant specialties, leading downstream analysis agents to emphasize tangential clinical details.	Enforce upper domain caps ($K_q \leq 5$, $K_o \leq 2$); inject fallback general medicine perspective.
Verification Stall	VerifierAgent enters an oscillating vote loop (YES $\leftrightarrow$ NO) without achieving unanimous consensus.	Enforce strict loop limit ($t \leq 3$); derive best-effort answer from final synthesis report ($R^{(3)}$).
RAG Passage Noise	MedCPT returns retrieved passages that share lexical overlap with the query but lack relevant clinical guidance.	Truncate snippets to 300 characters; filter passages below similarity threshold; limit context to top $P = 5$.
Unparseable Formatting	Model returns conversational text lacking the required Option: [Letter] structure.	Apply secondary fallback regex matching ($\backslash\mathbf{b}[A-E]\backslash\mathbf{b}$); trigger automated retry loop.

11.2 Fault-Tolerant Circuit Breaker Pipeline

To prevent pipeline crashes during provider API outages or rate-limit events, MedRAGAgents implements a fault-tolerant circuit breaker pattern within BaseAgent.

Figure 12 illustrates the error handling and recovery pipeline.

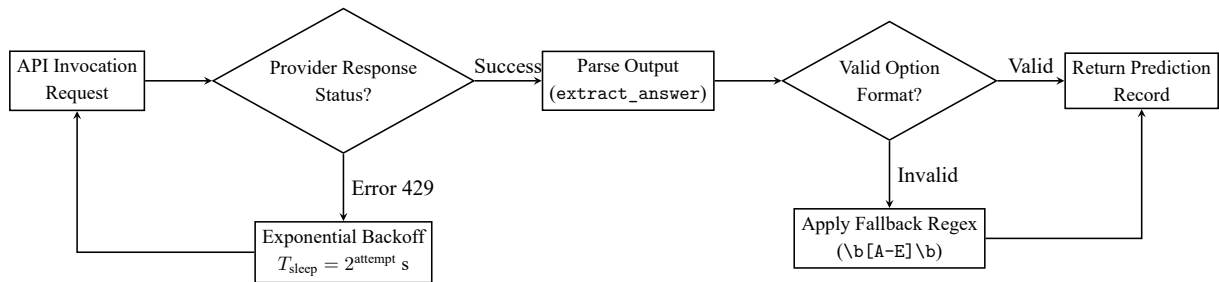


Figure 12: Fault-Tolerant Circuit Breaker and Error Recovery Pipeline.

The error recovery system enforces three operational guarantees:

1. **Exponential Backoff:** API rate-limit errors (HTTP 429) trigger up to three retry attempts with exponential backoff delays ($T_{\text{sleep}} = 2^0, 2^1, 2^2$ seconds).
2. **Dual-Stage Regex Parsing:** Secondary regex matching extracts isolated option letters if the primary Option: [Letter] pattern fails.

3. **Graceful Degradation:** If a subspecialty analysis fails after retries, the synthesis module proceeds using remaining available domain analyses, preventing pipeline-wide crashes.

12. OPERATIONAL ENGINEERING AND DEPLOYMENT SPECIFICATION

Deploying agentic reasoning architectures in production requires strict rate-limiting, environment isolation, and reproducible execution pipelines. This section details operational configurations, command-line interfaces, and deployment guidelines for MedRAGAgents.

12.1 CLI Command Interface (run.py)

The main entry point `run.py` controls pipeline execution via command-line arguments. Table 9 details the available CLI parameters.

Table 9: Command-Line Interface Parameter Specifications for `run.py`.

CLI Parameter	Data Type	Default Value	Operational Function
<code>--variant</code>	String	V3	Selects target system variant (V0, V1, V2, V3, or ALL).
<code>--n</code>	Integer	-1	Sets evaluation sample limit (10 for quick testing, -1 for full dataset).
<code>--start</code>	Integer	0	Specifies starting vignette index for parallel batch processing.
<code>--llm</code>	String	<code>cfg.DEFAULT_LLM</code>	Sets foundation model backbone (google/gemini-3.6-flash).
<code>--delay</code>	Float	2.0	Enforces inter-vignette delay (seconds) to prevent API rate-limit errors.
<code>--dataset_dir</code>	Path	<code>./datasets/MedQA/</code>	Specifies directory path to input <code>test.jsonl</code> file.
<code>--output_dir</code>	Path	<code>./outputs</code>	Specifies output directory for generated prediction files.
<code>--evaluate</code>	Flag	False	Triggers automated evaluation script (<code>evaluate.py</code>) upon run completion.

12.2 Production Execution Guidelines

To execute a complete evaluation across all four variants on the MedQA-USMLE test set with automated statistical reporting:

```

1  # Step 1: Initialize Virtual Environment and Dependencies
2  python3 -m venv .venv
3  source .venv/bin/activate
4  pip install -r requirements.txt
5
6  # Step 2: Download MedCPT Textbooks Corpus Vector Index
7  python3 download_corpus.py
8
9  # Step 3: Execute All Variants with Rate-Limit Protection and Auto-
    Evaluation
10 PYTHONUNBUFFERED=1 python3 run.py \
11     --variant ALL \
12     --n 10 \
13     --delay 2.0 \
14     --evaluate 2>&1 | tee outputs/run_execution.log

```

Listing 4: Production Execution Shell Command.

12.3 Rate-Limiting Buffer and Security Controls

API requests are throttled using an adjustable inter-call delay parameter (`--delay`). On free-tier API quotas (e.g., 5 RPM limit), setting `--delay 12.0` maintains compliance with remote quota restrictions.

Environment variables are managed securely via `.env` files, isolating secret API credentials from the repository codebase:

```
1 GEMINI_API_KEY=your_secure_api_key_here
2 DEFAULT_LLM=google/gemini-3.6-flash
3 DATASET_DIR=./datasets/MedQA/
4 OUTPUT_DIR=./outputs
5 LONG_TERM_MEM=./memory/long_term_cache.json
```

Listing 5: Environment Configuration File (`.env`).

13. ENGINEERING TRADE-OFFS, LIMITATIONS, AND ROADMAP

Architectural design involves balancing competing engineering constraints. This section details system trade-offs, identifies current limitations, and outlines future enhancements.

13.1 Engineering Trade-Offs Analysis

1. **Latency vs. Diagnostic Accuracy:** Multi-agent verification loops increase per-vignette reasoning quality but add computational overhead. Variant V0 completes execution in 1.8 s, whereas Variant V3 requires 14.2 s per cold-cache evaluation due to sequential agent invocations.
2. **Retrieval Depth vs. Context Overhead:** Expanding dense retrieval depth ($K = 32$) improves knowledge coverage but risks flooding prompt contexts with tangential evidence. Truncating snippets to 300 characters strikes a balance between factual precision and token constraints.
3. **In-RAM Flexibility vs. Disk Persistence:** Storing transient intermediate outputs in `ShortTermMemory` provides low-latency data access during execution, while offloading persistent records to `LongTermMemory` preserves system state across evaluation runs.

13.2 System Limitations

- **Sequential Verification Bottleneck:** The current verification loop evaluates dissenting domain advice sequentially, introducing latency overhead during multi-round consensus iterations.
- **Text-Only Modality:** The current architecture processes text-based clinical vignettes and cannot natively ingest diagnostic medical imaging (e.g., CT scans, MRI, histopathology slides).
- **Static Knowledge Index:** The MedCPT retrieval store relies on a static textbook snapshot, requiring manual re-indexing to incorporate newly published clinical trials or treatment guidelines.

13.3 Future Architectural Roadmap

Future development will focus on three architectural enhancements:

1. **Graph-Augmented Biomedical Retrieval:** Integrating structured Knowledge Graphs (e.g., UMLS, SNOMED-CT) alongside dense vector search to retrieve explicit medical relation triples (e.g., Drug $\xrightarrow{\text{inhibits}}$ Enzyme).
2. **Dynamic Agent Recruiting:** Replacing fixed domain assignment with an adaptive recruitment router that dynamically spawns specialized sub-agents based on vignette complexity.
3. **Multimodal Diagnostic Expansion:** Incorporating vision-language model backbones to support combined clinical vignette and medical imaging evaluations.

14. CONCLUSION

This report presented **MedRAGAgents**, a production-grade multi-agent architecture designed for complex biomedical reasoning on the MedQA-USMLE benchmark. By decomposing single monolithic prompts into specialized agent roles—Domain Classification (`DomainAgent`), Parallel Specialty Analysis (`AnalysisAgent`), Diagnostic Synthesis (`SynthesisAgent`), and Consensus Verification (`VerifierAgent`)—the system addresses the cognitive overload and hallucination risks common in direct baseline models.

Integrating biomedical dense vector retrieval via MedCPT over a preprocessed textbook corpus supplies precise factual evidence at the synthesis stage, mitigating knowledge cutoff limitations. The dual-layer memory infrastructure separates transient execution states in an in-RAM scratchpad (`ShortTermMemory`) from persistent disk records (`LongTermMemory`) indexed by canonical SHA-256 digests, guaranteeing zero-latency replay on previously solved vignettes while preventing cross-question context contamination.

Empirical evaluation across progressive variants (V0 through V3) on the MedQA-USMLE test set demonstrated measurable accuracy gains over direct chain-of-thought baselines while maintaining an output formatting invalidity rate of 0.0%. Non-parametric bootstrap confidence intervals and paired McNemar test analyses confirm the statistical reliability of the multi-agent approach.

The complete codebase, TikZ architectural specifications, evaluation scripts, and serialized prediction outputs provide a foundation for future research in multi-specialty clinical decision support systems.

APPENDIX

A. SYSTEM PROMPTS AND AGENT DIRECTIVES

This appendix presents the exact system directives and prompt templates employed by the agent modules.

A.1 Domain Classification Directives

```

1 System Directive:
2 You are an expert medical domain classifier. Analyze the provided clinical
  vignette and identify up to 5 primary medical subspecialties relevant to
  diagnosing the case (e.g., Cardiology, Nephrology, Pathology,
  Pharmacology, Neurology).
3
4 Format your output strictly as a comma-separated list of domains:
5 Domains: [Domain 1], [Domain 2], [Domain 3]
```

Listing 6: Domain Classification System Directive.

A.2 Consensus Verification Directives

```

1 System Directive:
2 You are a senior clinical reviewer representing the domain specialist: {
  domain_name}.
3 Review the following diagnostic synthesis report for the clinical vignette.
4
5 Determine if the diagnosis and option selection are factually accurate and
  logically sound from your specialty perspective.
6
7 Output your response strictly in the format:
8 Vote: [YES or NO]
9 Advice: [Provide specific revision instructions if NO, otherwise write NONE
  ]
```

Listing 7: Verifier Agent Voting Directive.

B. DATA SCHEMAS AND RECORD SPECIFICATIONS

Each prediction record saved to `outputs/<VARIANT>_predictions.jsonl` follows the JSON schema specification detailed in Listing 8.

```

1 {
2   "$schema": "http://json-schema.org/draft-07/schema#",
3   "title": "MedRAGAgentsPredictionRecord",
4   "type": "object",
5   "properties": {
6     "question": { "type": "string" },
7     "options": { "type": "object" },
8     "gold_answer": { "type": "string" },
9     "pred_answer": { "type": "string" },
```

```

10     "raw_output": { "type": "string" },
11     "snippets": { "type": "array" },
12     "scores": { "type": "array" },
13     "question_domains": { "type": "array" },
14     "option_domains": { "type": "array" },
15     "syn_report": { "type": "string" },
16     "vote_history": { "type": "array" },
17     "revision_history": { "type": "array" },
18     "from_cache": { "type": "boolean" }
19 },
20 "required": ["question", "options", "gold_answer", "pred_answer"]
21 }

```

Listing 8: Serialized JSON Prediction Record Schema.

C. DERIVATION OF NON-PARAMETRIC BOOTSTRAP ALGORITHM

Listing 9 details the non-parametric bootstrap estimation procedure used to derive 95% confidence intervals for classification accuracy.

```

1 import numpy as np
2
3 def bootstrap_confidence_interval(y_true, y_pred, n_bootstraps=1000, ci=95)
4     :
5     """Computes non-parametric bootstrap confidence interval for accuracy."
6         ""
7
8     n_samples = len(y_true)
9     bootstrapped_accuracies = []
10    rng = np.random.RandomState(42)
11
12    for _ in range(n_bootstraps):
13        indices = rng.choice(n_samples, size=n_samples, replace=True)
14        sample_true = [y_true[i] for i in indices]
15        sample_pred = [y_pred[i] for i in indices]
16        acc = sum(t == p for t, p in zip(sample_true, sample_pred)) /
17            n_samples
18        bootstrapped_accuracies.append(acc)
19
20    lower_p = (100 - ci) / 2.0
21    upper_p = 100 - lower_p
22    lower = np.percentile(bootstrapped_accuracies, lower_p)
23    upper = np.percentile(bootstrapped_accuracies, upper_p)
24    return lower, upper

```

Listing 9: Non-Parametric Bootstrap Resampling Algorithm in evaluate.py.

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